EXPERIMENT – 13

EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS AND STORE THE 32-BIT RESULT IN INTERNAL RAM

AIM: To write an Embedded C program to add an array of 16-bit numbers and store the result in internal RAM using Keil software.

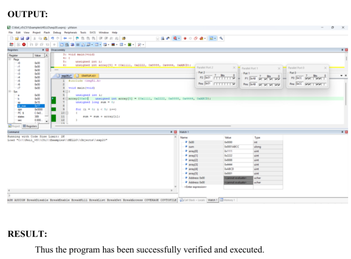
OBJECTIVES:

1. To store an array of 16-bit hexadecimal numbers.
2. To calculate the sum of all array elements.
3. To store the 32-bit result across the microcontroller ports.

PROGRAM:

```
#include <reg51.h>
void main(void)
{
    unsigned int i;
    unsigned int array[5] = {0x1111,0x2222,0x8888,0x4444,0xABCD};
    unsigned long sum = 0;
    for(i=0;i<5;i++)
    {
        sum = sum + array[i];
    }
    P0 = (unsigned char)(sum & 0xFF);
    P1 = (unsigned char)((sum >> 8) & 0xFF);
    P2 = (unsigned char)((sum >> 16) & 0xFF);
    while(1);
}
```

OUTPUT IMAGE:



RESULT:

Thus the program has been successfully verified and executed.